

GREENVILLE-SPARTANBURG, SC, USA

WMO: 723120

Lat: 34.906N

Lon: 82.213W

Elev: 291

StdP: 97.88

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-3.7	-16.5	0.9	-1.0	-14.2	1.1	0.4	10.1	9.9	8.9	9.9	2.3	340	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.5	23.0	33.2	22.9	32.0	22.6	25.0	30.8	24.4	29.9	23.9	29.2	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.8	26.7	22.9	18.2	26.2	22.4	17.7	25.9	77.0	30.8	75.0	30.2	72.9	29.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.5	6.7	-9.7	36.3	2.3	2.1	-11.4	37.8	-12.8	39.0	-14.1	40.2	-15.8	41.7		
			-11.1	26.0	2.0	0.9	-12.5	26.6	-13.7	27.1	-14.8	27.6	-16.3	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	5.8	7.9	11.7	16.3	20.7	24.7	26.3	25.8	22.6	16.7	11.1	7.4
	DBStd	8.24	5.28	4.88	4.92	4.06	3.42	2.53	2.05	2.16	3.07	4.05	4.39	4.76
	HDD10.0	437	151	91	40	4	0	0	0	0	0	3	39	110
	HDD18.3	1639	388	293	211	84	19	1	0	5	78	220	340	
	CDD10.0	2795	21	31	93	194	331	441	506	489	378	211	71	29
	CDD18.3	954	0	1	5	24	91	192	247	231	133	28	2	1
	CDH23.3	8117	1	3	47	237	744	1679	2264	1983	954	192	12	1
	CDH26.7	3168	0	0	6	42	218	693	1009	830	334	36	0	0
Wind	WSAvg	2.8	3.1	3.2	3.4	3.4	2.9	2.6	2.4	2.2	2.4	2.5	2.6	2.7
Precipitation	PrecAvg	1275	104	106	133	95	104	115	120	110	100	96	91	102
	PrecMax	1788	183	212	289	287	226	257	345	441	296	260	199	256
	PrecMin	890	7	14	29	18	28	4	19	20	4	0	14	9
	PrecStd	221	42	50	63	60	54	62	69	78	70	67	38	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	23.4	27.4	29.7	32.8	35.3	36.1	36.6	33.7	29.6	25.1	22.2
		MCWB	15.1	15.9	16.7	18.6	21.4	22.8	23.6	23.1	21.2	20.7	16.7	16.2
	2%	DB	18.5	20.5	24.5	27.9	30.7	33.6	34.4	34.2	32.0	27.4	22.5	19.6
		MCWB	14.0	13.4	15.4	17.9	20.7	22.6	23.6	23.0	21.4	19.1	15.6	15.3
	5%	DB	16.5	18.3	22.4	26.2	29.2	32.2	33.1	32.4	30.3	25.6	20.6	17.3
		MCWB	12.5	12.7	14.2	17.0	20.5	22.4	23.4	23.1	21.4	18.3	15.1	13.8
	10%	DB	14.0	16.0	20.1	24.3	27.6	30.6	31.7	30.9	28.6	23.5	18.5	15.1
		MCWB	10.6	11.0	13.0	16.0	19.6	21.9	23.2	22.9	20.9	17.7	14.1	11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.9	18.4	20.7	23.4	25.0	25.7	25.7	24.1	23.2	20.4	18.8
		MCDB	19.4	21.0	22.9	26.7	28.8	31.3	32.2	31.7	28.9	26.4	21.6	20.1
	2%	WB	15.7	16.3	17.1	19.7	22.4	24.1	25.0	25.0	23.5	21.6	18.0	16.6
		MCDB	17.6	19.0	21.8	25.2	27.8	30.3	31.3	30.4	27.8	24.7	20.2	18.5
	5%	WB	13.5	14.2	15.9	18.6	21.6	23.6	24.5	24.4	22.8	20.3	16.7	14.7
		MCDB	15.7	16.9	20.3	23.2	26.7	29.4	30.4	29.4	26.9	23.4	19.1	16.8
	10%	WB	11.2	12.0	14.7	17.5	20.8	22.9	24.0	23.8	22.1	19.0	15.1	12.3
		MCDB	13.0	14.6	18.6	22.0	25.7	28.3	29.5	28.5	26.1	22.0	17.8	14.0
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.8	11.6	12.0	11.3	10.7	10.3	9.9	10.0	11.0	11.1	9.7
		MCDBR	12.3	13.2	14.1	14.1	12.9	12.7	12.2	12.0	12.2	12.7	13.3	11.5
		MCWBR	8.3	8.4	7.5	6.3	5.2	4.2	3.7	3.7	4.2	5.8	7.4	7.5
	5% WB	MCDBR	10.0	10.8	11.9	11.6	10.8	10.8	10.5	9.9	9.4	9.6	9.7	9.4
		MCWBR	7.9	8.3	7.5	6.1	4.9	4.2	3.8	3.7	4.0	5.4	7.4	7.4
Clear-Sky Solar Irradiance	taub		0.310	0.320	0.354	0.394	0.452	0.479	0.525	0.513	0.416	0.379	0.333	0.308
	taud		2.509	2.479	2.412	2.298	2.172	2.133	2.030	2.047	2.299	2.385	2.473	2.545
	Ebn at Noon		899	928	918	891	841	814	774	775	842	847	861	881
	Edh at Noon		83	95	110	130	149	155	171	165	122	103	85	76
All-Sky Solar Radiation	RadAvg		2.66	3.29	4.36	5.43	6.00	6.33	5.93	5.48	4.71	3.91	2.95	2.29
	RadStd		0.15	0.37	0.29	0.40	0.47	0.45	0.39	0.26	0.42	0.44	0.26	0.22

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+100
(47)	Regional (54 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+113

Nomenclature: See separate page